

Hive Robotics

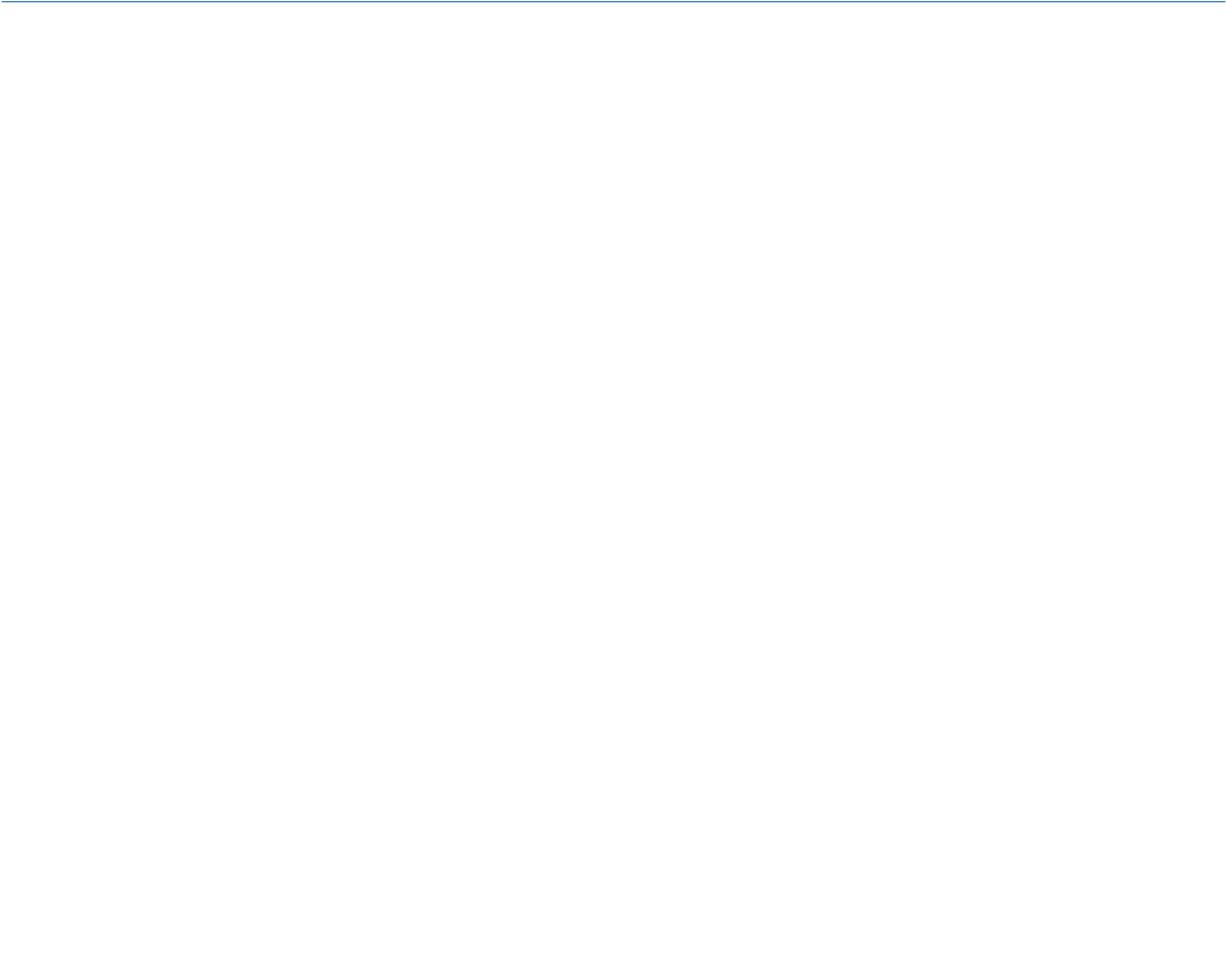
Complete System Architecture · v1.0

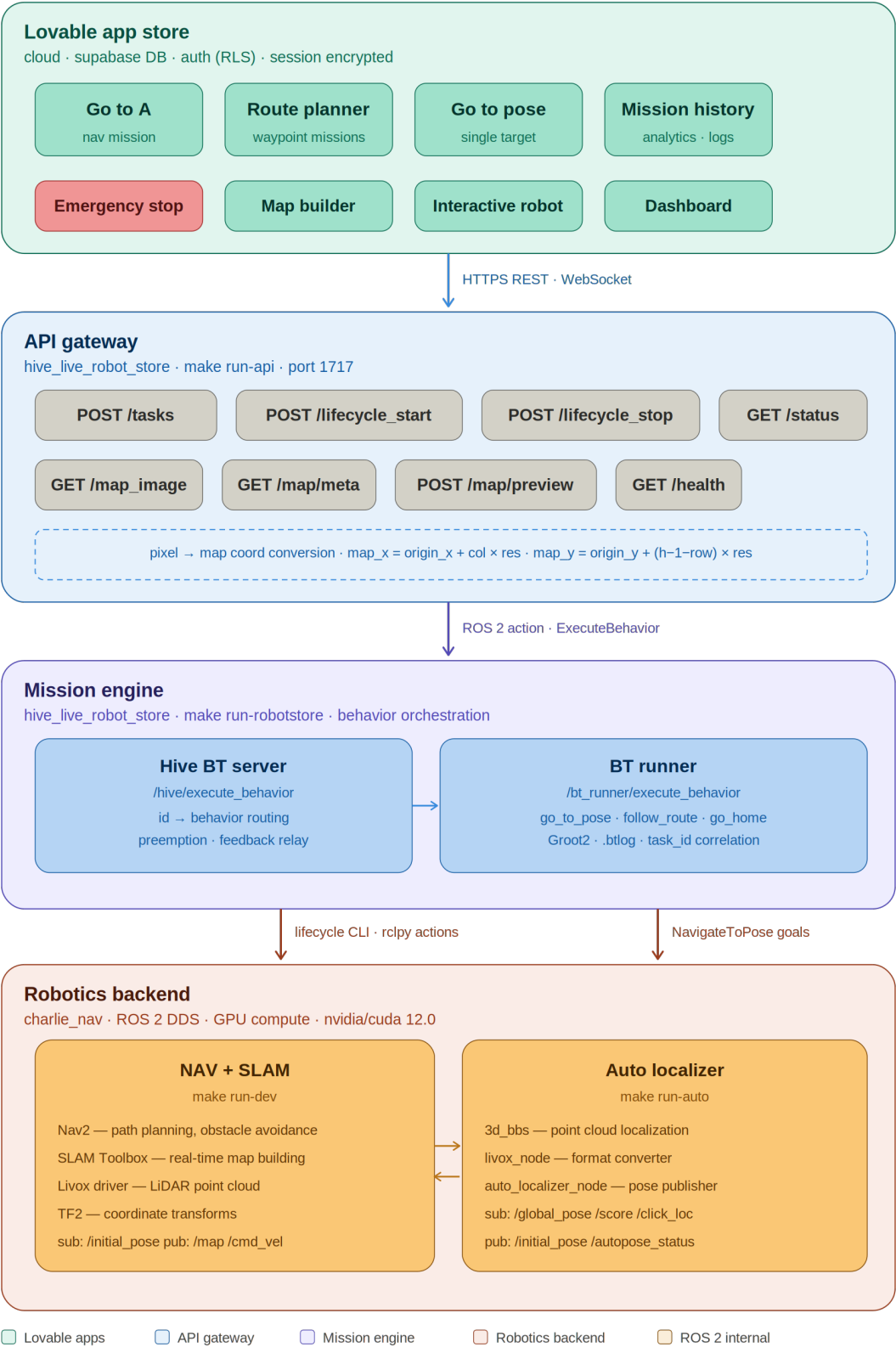
ROS 2 Humble · FastAPI · Docker · Lovable

This document describes the four-layer architecture of the Hive autonomous robot platform. Each layer has a single responsibility and communicates with adjacent layers through well-defined interfaces. No layer skips another. The diagram below shows the complete signal flow from the Lovable web app down to the physical robot hardware.

Layer	Name	Repo	Key Tech
1 (top)	Lovable app store	lovable.dev	React · Supabase · Auth RLS
2	API gateway	hive_live_robot_store	FastAPI · Port 1717 · Uvicorn
3	Mission engine	hive_live_robot_store	BehaviorTree.CPP · Groot2
4 (bot)	Robotics backend	charlie_nav	ROS 2 · Nav2 · SLAM · CUDA

System Architecture Diagram





Key Interface Notes

Interface	Protocol	Detail
Lovable → API gateway	HTTPS REST / WebSocket	Commands via REST, live telemetry via WebSocket

API → Mission engine	ROS 2 Action (ExecuteBehavior)	Pixel coords converted to map-frame metres by API
API → Robotics backend	lifecycle CLI + rclpy	lifecycle set activate/deactivate/configure/cleanup
Mission engine → Nav2	ROS 2 Action (NavigateToPose)	BT runner dispatches one goal per waypoint
Auto localizer → NAV+SLAM	ROS 2 Topic (/initial_pose)	Pose published after 3d_bbs localization converges
API → Auto localizer	ROS 2 Topic (/click_loc Bool)	Triggers re-localization from API endpoint

Docker Startup Order

Step	Command	Repo	Starts
1	make run-dev	charlie_nav	NAV + SLAM
2	make run-auto	charlie_nav	Auto localizer
3	make run-robotstore	hive_live_robot_store	Mission engine (BT)
4	make run-api	hive_live_robot_store	API gateway :1717